Static Electricity (Supplementary Problems)

1. Coulomb's Law in Two Dimensions Sphere A, with a charge of +6.0 μC, is located near another charged sphere, B. Sphere B has a charge of -3.0 μC and is located 4.0 cm to the right of A.
a. What is the force of sphere B on sphere A?
b. A third sphere, C, with a +1.5-μC charge, is added to the configuration. If it is located 3.0 cm directly beneath A, what is the new net force on sphere A?

2. A negative charge of -2.0×10^{-4} C and a positive charge of 8.0×10^{-4} C are separated by 0.30 m. What is the force between the two charges?

3. A negative charge of -6.0×10^{-6} C exerts an attractive force of 65 N on a second charge that is 0.050 m away. What is the magnitude of the second charge?

4. The charge on B in Problem 1 is replaced by a charge of +3.00 μC. Diagram the new situation and find the net force on A.

5. Sphere A is located at the origin and has a charge of $+2.0\times10^{-6}$ C. Sphere B is located at +0.60 m on the x-axis and has a charge of -3.6×10^{-6} C. Sphere C is located at +0.80 m on the x-axis and has a charge of $+4.0\times10^{-6}$ C. Determine the net force on sphere A.

6. Determine the net force on sphere B in the previous problem.

7. Two charges, q_A and q_B, are separated by a distance, r, and exert a force, F, on each other. Analyze Coulomb's law and identify what new force would exist under the following conditions.
a. q_A is doubled b. q_A and q_B are cut in half c. r is tripled d. r is cut in half e. q_A is tripled and r is doubled

8. Lightning A strong lightning bolt transfers about 25 C to Earth. How many electrons are transferred?

9. Atoms Two electrons in an atom are separated by 1.5×10^{-10} m, the typical size of an atom. What is the electric force between them?

10. A positive and a negative charge, each of magnitude 2.5×10^{-5} C, are separated by a distance of 15 cm. Find the force on each of the particles.

11. A force of 2.4×10^{2} N exists between a positive charge of 8.0×10^{-5} C and a positive charge of 3.0×10^{-5} C. What distance separates the charges?

12. Two identical positive charges exert a repulsive force of 6.4×10^{-9} N when separated by a distance of 3.8×10^{-10} m. Calculate the charge of each.

13. A positive charge of 3.0 μC is pulled on by two negative charges. As shown in Figure 20-14, one negative charge, -2.0 μC, is 0.050 m to the west, and the other, -4.0 μC, is 0.030 m to the east. What total force is exerted on the positive charge?

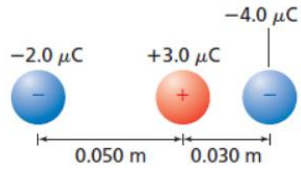


Figure 20-14

14. Figure 20-15 shows two positively charged spheres, one with three times the charge of the other. The spheres are 16 cm apart, and the force between them is 0.28 N. What are the charges on the two spheres?

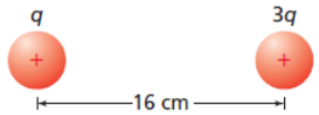


Figure 20-15

15. Charge in a Coin How many coulombs of charge are on the electrons in a nickel? Use the following method to find the answer.
a. Find the number of atoms in a nickel. A nickel has a mass of about 5 g. A nickel is 75 percent Cu and 25 percent Ni, so each mole of the coin's atoms will have a mass of about 62 g.
b. Find the number of electrons in the coin. On average, each atom has 28.75 electrons.
c. Find the coulombs on the electrons.

16. Three particles are placed in a line. The left particle has a charge of -55 μC, the middle one has a charge of -45 μC, and the right one has a charge of +78 μC. The middle particle is 72 cm from each of the others, as shown in Figure 20-16. a. Find the net force on the middle particle. b. Find the net force on the right particle.

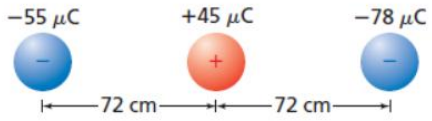


Figure 20-16

17. A small metal sphere with charge 1.2×10^{-5} C is touched to an identical neutral sphere and then placed 0.15 m from the second sphere. What is the electric force between the two spheres?

18. Atoms What is the electric force between an electron and a proton placed 5.3×10^{-11} m apart, the approximate radius of a hydrogen atom?

19. A small sphere of charge 2.4 μC experiences a force of 0.36 N when a second sphere of unknown charge is placed 5.5 cm from it. What is the charge of the second sphere?

20. Two identically charged spheres placed 12 cm apart have an electric force of 0.28 N between them. What is the charge of each sphere?

21. In an experiment using Coulomb's apparatus, a sphere with a charge of 3.6×10^{-8} C is 1.4 cm from a second sphere of unknown charge. The force between the spheres is 2.7×10^{-2} N. What is the charge of the second sphere?

22. The force between a proton and an electron is 3.5×10^{-10} N. What is the distance between these two particles?

23. Apply Concepts Calculate the ratio of the electric force to the gravitational force between the electron and the proton in a hydrogen atom.

24. Analyze and Conclude Sphere A, with a charge of +64 μC, is positioned at the origin. A second sphere, B, with a charge of -16 μC, is placed at +1.00 m on the x-axis.

a. Where must a third sphere, C, of charge +12 μC be placed so there is no net force on it?

b. If the third sphere had a charge of +6 μC, where should it be placed?

c. If the third sphere had a charge of -12 μC, where should it be placed?

25. Three charged spheres are located at the positions shown in Figure 20-17. Find the total force on sphere B.

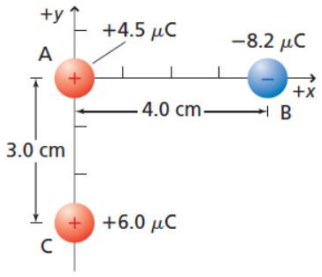


Figure 20-17

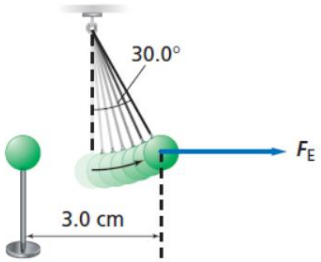


Figure 20-18

26. The two pith balls in Figure 20-18 each have a mass of 1.0 g and an equal charge. One pith ball is suspended by an insulating thread. The other is brought to 3.0 cm from the suspended ball. The suspended ball is now hanging with the thread forming an angle of 30.0° with the vertical. The ball is in equilibrium with FE, Fg, and FT. Calculate each of the following.

a. F_g on the suspended ball

b. F_E

c. the charge on the balls

27. Two charges, q_A and q_B, are at rest near a positive test charge, q_T, of 7.2 μC. The first charge, q_A, is a positive charge of 3.6 μC located 2.5 cm away from q_T at 35°; q_B is a negative charge of -6.6 μC located 6.8 cm away at 125°.

a. Determine the magnitude of each of the forces acting on q_T.

b. Sketch a force diagram.

c. Graphically determine the resultant force acting on q_T.

28. How many electrons have been removed from a positively charged electroscope if it has a net charge of 7.5×10^{-10} C?

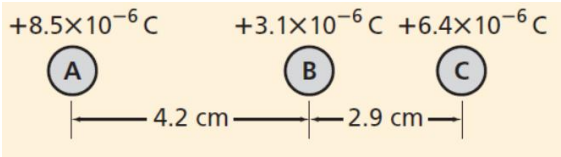
a) 4.7×10^{9} electrons b) 2.1×10^{9} electrons c) 1.2×10^{9} electrons d) 7.5×10^{9} electrons

29. The force exerted on a particle with a charge of 5.0×10^{-19} C by a second particle that is 4 cm away is 8.4×10^{-8} N. What is the charge of the second particle?

a) 4.2×10^{-13} C b) 3.0×10^{-9} C c) 6.0×10^{-9} C d) 8.4×10^{-13} C

30. Three charges, A, B, and C, are located in a line at known locations. What is the net force on charge B?

a) 78 N toward A
b) 78 N toward C
c) 130 N toward A
d) 210 N toward C



31. What is the charge on an electroscope that has an excess of 4.8×10^{10} electrons?

a) 3.3×10^{-9} C b) 7.7×10^{-9} C c) 4.8×10^{-9} C d) 4.8×10^{-10} C

32. Two charged bodies exert a force of 86 N on each other. If they are moved so that they are six times farther apart, what is the new force that they will exert on each other?

a) 24 N b) 14 N c) 86 N d) 5.2×10^{-9} N

33. Two equally charged bodies exert a force of 90 N on each other. If one of the bodies is exchanged for a body of the same size, but three times as much charge, what is the new force that they will exert on each other?

a) 10 N b) 30 N c) 2.7×10^{2} N d) 8.1×10^{2} N

34. An alpha particle has a mass of 6.68×10^{-27} kg and a charge of 3.2×10^{-19} C. What is the ratio of the electrostatic force to the gravitational force between two alpha particles?

a) 1 b) 4.8×10^{8} c) 2.3×10^{15} d) 4.8×10^{18}

35. Charging a neutral body by touching it with a charged body is called:

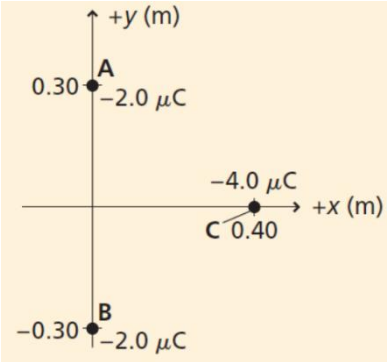
a) conduction b) induction c) grounding d) discharging

36. Macys uses helium balloons with wool, giving the balloon a charge of -8.9×10^-14 C. What is the force between the balloon and a metal sphere that is charged to 25 C and is 2 km away?

a) 8.9×10^{-5} N b) 2.2×10^{-12} N c) 5.0×10^{-9} N d) 5.6×10^{-4} N

37. According to the diagram, what is the net force exerted by charges A and B on charge C? In your answer, include a diagram showing the force vectors F_A on C, F_B on C, and F_{net}.

a) +y (m)
b) -y (m)
c) +x (m)
d) -x (m)



38. What happens to the speed of a wave on a string when the frequency is doubled? Assume the tension in the string remains the same.

39. How many electrons have been removed from a positively charged electroscope if it has a net charge of 7.5×10^{-11} C?

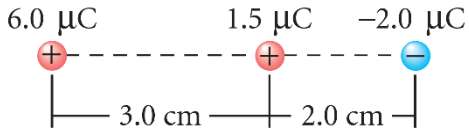
40. Three-point charges, q_1, q_2, and q_3, lie along the x-axis at $x = 0$, $x = +3.0$ cm, and $x = +5.0$ cm, respectively. Calculate the magnitude and direction of the net force on the negative-point charge when $q_1 = +6.0$ μC, $q_2 = +1.5$ μC, and $q_3 = -2.0$ μC.

41. (20pts) A charge q_1 of -6.00×10^{-9} C and a charge q_2 of -3.00×10^{-9} C are separated by a distance of 60.0 cm. Where could a third charge be placed so that the net electric force on it is zero?

42. (40pts) Consider three charges arranged as shown below.

a. What is the electric field strength at a point 1.0 cm to the left of the middle charge?

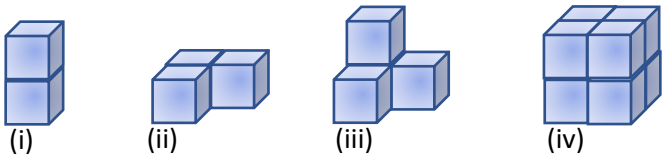
b. What is the magnitude of the force on a 2.0 μC charge placed at this point?



43. (40pts) Two-point charges are located on the positive x axis of a coordinate system. Charge $q_1 = 3.0$ nC is 2.0 cm from the origin, and charge $q_2 = -7.0$ nC is 4.0 cm from the origin. What is the total force (magnitude and direction) exerted by these two charges on a third point charge $q_3 = 5.0$ nC located at the origin?

44. (20pts) Solid plastic cubes, each 3.00 cm on each edge, are glued together to form each one of the objects (i, ii, iii, and iv) shown in figure below. Assuming one small cube carries 1nC of electric charge. How many excess electrons does each object have?





45. Two identical positive charges exert a repulsive force of 6.4×10^{9} N when separated by a distance of 3.8×10^{-10} m. Calculate the charge of each.

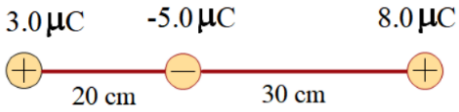
46. The electron and proton of a hydrogen atom are separated (on the average) by a distance of about 5.3×10^{-11}.

a) Find the magnitudes of the electric force.

b) the gravitational force that each particle exerts on the others.

c) the ratio of the electric force F_e to the gravitational force Fg.

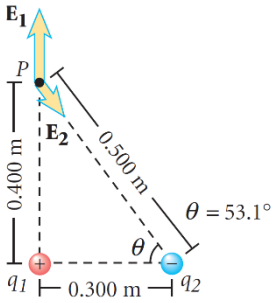
47. Three-point charges are placed on the axis as shown in figure below. Find the net force on the -5.0μC due to the two other charges.



48. The force exerted on a particle with a charge of $5.0\times10^{-9}C$ by a second particles that is 4 cm away is $8.4\times10^{-5}N$ .What is the charge of the second particles?

49. Three charges lie along the x-axis. One positive charge, q_1 = 15 μC, is at x = 2.0 m, and another positive charge, q_2 = 6.0 μC, is at the origin. At what point on the x-axis must a negative charge, q_3, be placed so that the resultant force on it is zero?

50. A charge $q_1 = +7.00\ \mu C$ is at the origin, and a charge $q_2 = -5.00\ \mu C$ is on the x-axis 0.300 m from the origin, as shown at right. Find the electric field strength at point P, which is on the y-axis 0.400 m from the origin.



51. An electric field of 2.0×10^4 N/C is directed along the positive x-axis.

a. What is the electric force on an electron in this field?

b. What is the electric force on a proton in this field?

Electric Field (Supplementary Problems)

52. A positive test charge of 5.0×10^{-6} C is in an electric field that exerts a force of 2.0×10^{-4} N on it. What is the magnitude of the electric field at the location of the test charge?

53. A negative charge of 2.0×10^{-8} C experiences a force of 0.060 N to the right in an electric field. What are the field's magnitude and direction at that location?

54. A positive charge of 3.0×10^{-7} C is located in a field of 27 N/C directed toward the south. What is the force acting on the charge?

55. A pith ball weighing 2.1×10^{-3} N is placed in a downward electric field of 6.5×10^{4} N/C. What charge (magnitude and sign) must be placed on the pith ball so that the electric force acting on it will suspend it against the force of gravity?

56. You are probing the electric field of a charge of unknown magnitude and sign. You first map the field with a 1.0×10^{-6}-C test charge, then repeat your work with a 2.0×10^{-6}-C test charge. a. Would you measure the same forces at the same place with the two test charges? Explain. b. Would you find the same field strengths? Explain.

57. What is the magnitude of the electric field strength at a position that is 1.2 m from a point charge of 4.2×10^{-6} C?

58. What is the magnitude of the electric field strength at a distance twice as far from the point charge in problem 6?

59. What is the electric field at a position that is 1.6 m east of a point charge of $+7.2\times10^{-6}$ C?

60. The electric field that is 0.25 m from a small sphere is 450 N/C toward the sphere. What is the charge on the sphere?

61. How far from a point charge of $+2.4\times10^{-6}$ C must a test charge be placed to measure a field of 360 N/C?

62. The electric field intensity between two large, charged, parallel metal plates is 6000 N/C. The plates are 0.05 m apart. What is the electric potential difference between them?

63. A voltmeter reads 400 V across two charged, parallel plates that are 0.020 m apart. What is the electric field between them?

64. What electric potential difference is applied to two metal plates that are 0.200 m apart if the electric field between them is 2.50×10^3 N/C?

65. When a potential difference of 125 V is applied to two parallel plates, the field between them is 4.25×10^3 N/C. How far apart are the plates?

66. A potential difference of 275 V is applied to two parallel plates that are 0.35 cm apart. What is the electric field between the plates?

67. What work is done when 3.0 C is moved through an electric potential difference of 1.5 V?

68. A 12-V car battery can store 1.44×10^{6} C when it is fully charged. How much work can be done by this battery before it needs recharging?

69. An electron in a television picture tube passes through a potential difference of 18,000 V. How much work is done on the electron as it passes through that potential difference?

70. If the potential difference in problem 18 is between two parallel plates that are 2.4 cm apart, what is the magnitude of the electric field between them?

71. The electric field in a particle-accelerator machine is 4.5×10^5 N/C. How much work is done to move a proton 25 cm through that field?

72. A drop is falling in a Millikan oil-drop apparatus with no electric field. What forces are acting on the oil drop, regardless of its acceleration? If the drop is falling at a constant velocity, describe the forces acting on it.

73. An oil drop weighs 1.9×10^{-15} N. It is suspended in an electric field of 6.0×10^{3} N/C. What is the charge on the drop? How many excess electrons does it carry?

74. An oil drop carries one excess electron and weighs 6.4×10^{-15} N. What electric field strength is required to suspend the drop so it is motionless?

75. A positively charged oil drop weighing 1.2×10^{-14} N is suspended between parallel plates separated by 0.64 cm. The potential difference between the plates is 240 V. What is the charge on the drop? How many electrons is the drop missing?

76. A 27-μF capacitor has an electric potential difference of 45 V across it. What is the charge on the capacitor?

77. Both a 3.3-μF and a 6.8-μF capacitor are connected across a 24-V electric potential difference. Which capacitor has a greater charge? What is it?

78. The same two capacitors as in problem 26 are each charged to 3.5×10^{-4} C. Which has the larger electric potential difference across it? What is it?

79. A 2.2-μF capacitor first is charged so that the electric potential difference is 6.0 V. How much additional charge is needed to increase the electric potential difference to 15.0 V?

80. When a charge of 2.5×10^{-5} C is added to a capacitor, the potential difference increases from 12.0 V to 14.5 V. What is the capacitance of the capacitor?

81. What charge exists on a test charge that experiences a force of 1.4×10^{-8} N at a point where the electric field intensity is 5.0×10^{-4} N/C?

82. A positive charge of 1.0×10^{-5} C, shown in Figure 21-19, experiences a force of 0.30 N when it is located at a certain point. What is the electric field intensity at that point?

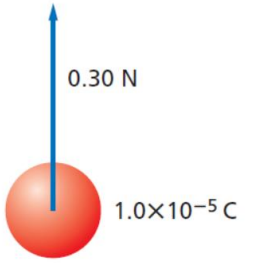


Figure 21-19

83. A test charge experiences a force of 0.30 N on it when it is placed in an electric field intensity of

4.5×10^5 N/C. What is the magnitude of the charge?

84. The electric field in the atmosphere is about 150 N/C downward. a. What is the direction of the force on a negatively charged particle? b. Find the electric force on an electron with charge -1.6×10^{-19} C. c. Compare the force in part b with the force of gravity on the same electron (mass = 9.1×10^{-31} kg).

85. Carefully sketch each of the following. a. the electric field produced by a +1.0-μC charge b. the electric field resulting from a +2.0-μC charge (Make the number of field lines proportional to the change in

charge.)

86. A positive test charge of 6.0×10^{-6} C is placed in an electric field of 50.0-N/C intensity, as in Figure 21-20. What is the strength of the force exerted on the test charge?

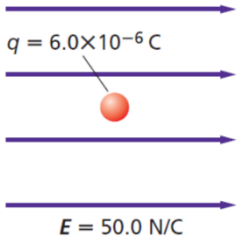


Figure 21-20

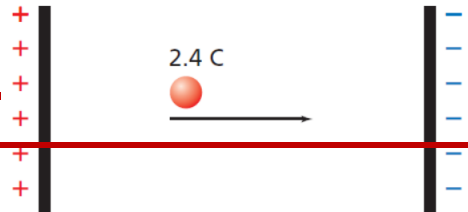
87. Charges X, Y, and Z all are equidistant from each other. X has a +1.0-μC charge, Y has a +2.0-μC charge, and Z has a small negative charge. a. Draw an arrow representing the force on charge Z. b. Charge Z now has a small positive charge on it. Draw an arrow representing the force on it.

88. In a television picture tube, electrons are accelerated by an electric field having a value of 1.00×10^{5} N/C. a. Find the force on an electron. b. If the field is constant, find the acceleration of the electron (mass = 9.11×10^{-31} kg).

89. What is the electric field strength 20.0 cm from a point charge of 8.0×10^{-7} C?

90. The nucleus of a lead atom has a charge of 82 protons. a. What are the direction and magnitude of the electric field at 1.0×10^{-10} m from the nucleus? b. What are the direction and magnitude of the force exerted on an electron located at this distance?

91. If 120 J of work is performed to move 2.4 C of charge from the positive plate to the negative plate shown in Figure 21-21, what potential difference exists between the plates?



92. How much work is done to transfer 0.15 C of charge through an electric potential difference of 9.0 V?

93. An electron is moved through an electric potential difference of 450 V. How much work is done on the electron?

94. A 12-V battery does 1200 J of work transferring charge. How much charge is transferred?

95. The electric field intensity between two charged plates is 1.5×10^3 N/C. The plates are 0.060 m apart. What is the electric potential difference, in volts, between the plates?

96. A voltmeter indicates that the electric potential difference between two plates is 70.0 V. The plates are 0.020 m apart. What electric field intensity exists between them?

97. A capacitor that is connected to a 45.0-V source contains 90.0 μC of charge. What is the capacitor's capacitance?

98. What electric potential difference exists across a 5.4-μF capacitor that has a charge of 8.1×10^{-4} C?

99. The oil drop shown in Figure 21-22 is negatively charged and weighs 4.5×10^{-15} N. The drop is suspended in an electric field intensity of 5.6×10^{3} N/C. a. What is the charge on the drop? b. How many excess electrons does it carry?

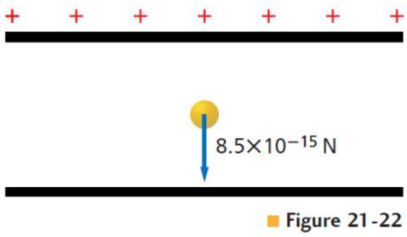


Figure 21-22

100. What is the charge on a 15.0-pF capacitor when it is connected across a 45.0-V source?

101. A force of 0.065 N is required to move a charge of 37 μC a distance of 25 cm in a uniform electric field, as in Figure 21-23. Wh at is the size of the electric potential difference between the two points?

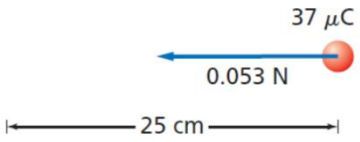


Figure 21-23

102. Photoflash The energy stored in a capacitor with capacitance C, and an electric potential difference, ΔV, is represented by $W = 1/2C\Delta V^2$. One application of this is in the electronic photoflash of a strobe light, like the one in Figure 21-24. In such a unit, a capacitor of 10.0 μF is charged to 3.0×10^2 V. Find the energy stored



Figure 21-24

103. Suppose it took 25 s to charge the capacitor in problem 51. a. Find the average power required to charge the capacitor in this time. b. When this capacitor is discharged through the strobe lamp, it transfers all its energy in 1.0×10^{-4} s. Find the power delivered to the lamp. c. How is such a large amount of power possible?

104. Lasers Lasers are used to try to produce controlled fusion reactions. These lasers require brief pulses of energy that are stored in large rooms filled with capacitors. One such room has a capacitance of 61×10^{-3} F charged to a potential difference of 10.0 kV. a. Given that $W = 1/2\ C\Delta V^2$, find the energy stored in the capacitors. b. The capacitors are discharged in 10 ns (1.0×10^{-8} s). What power is produced? c. If the capacitors are charged by a generator with a power capacity of 1.0 kW, how many seconds will be required to charge the capacitors?

105. How much work does it take to move 0.25 μC between two parallel plates that are 0.40 cm apart if the field between the plates is 6400 N/C?

106. How much charge is stored on a 0.22-μF parallel plate capacitor if the plates are 1.2 cm apart and the electric field between them is 2400 N/C?

107. Two identical small spheres, 25 cm apart, carry equal but opposite charges of 0.060 μC, as in Figure 21-25. If the potential difference between them is 300 V, what is the capacitance of the system?

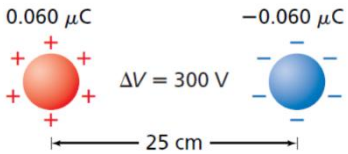


Figure 21-25

108. The plates of a 0.047 μF capacitor are 0.25 cm apart and are charged to a potential difference of 120 V. How much charge is stored on the capacitor?

109. What is the strength of the electric field between the plates of the capacitor in Problem 57 above?

110. An electron is placed between the plates of the capacitor in Problem 58 above, as in Figure 21-26. What force is exerted on that electron?

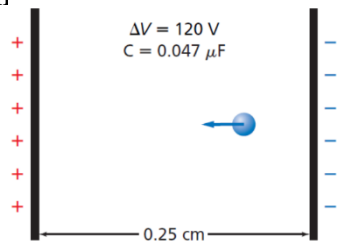


Figure 21-26

111. Why is an electric field measured only by a small test charge?

a) so the charge doesn't disturb the field
b) because small charges have small momentum
c) so its size doesn't nudge the charge to be measured aside
d) because an electron always is used as the test charge and electrons are small

112. A force of 14 N exists on a charge q, which is 2.1×10^{-8} C. What is the magnitude of the electric field? a) 0.15×10^{-9} N/C b) 6.7×10^{-9} N/C c) 29×10^{-9} N/C d) 6.7×10^{-9} N/C

113. A positive test charge of 8.7 μC experiences a force of 8.1×10^-6 N at an angle of 24° N of E. What are the magnitude and direction of the electric field strength at the location of the test charge?
a) 7.0×10^{-8} N/C, 24° N of E b) 1.7×10^{-8} N/C, 24° S of W c) 1.1×10^{-8} N/C, 24° W of S d) 9.3×10^{-8} N/C, 24° N of E

114. What is the potential difference between two places that are 18 cm apart with a field of 4.8×10^{-4} N/C? a) 27 V b) 86 V c) 0.86 kV d) 27 kV

115. How much work is done on a proton to move it from the negative plate to a positive plate 4.3 cm away if the field is 125 N/C?

a) 5.5×10^{-23} J b) 8.6×10^{-19} J c) 1.1×10^{-16} J d) 5.4 J

116. How was the magnitude of the field in Millikan's oil-drop experiment determined?

a) using a measurable electromagnet
b) from the electric potential between the plates
c) from the magnitude of the charge
d) by an electrometer

117. In an oil drop experiment, a drop with a weight of 1.9×10^{-14} N was suspended when the potential difference between the plates that were 6.3 mm apart was 0.78 kV. What was the charge on the drop?

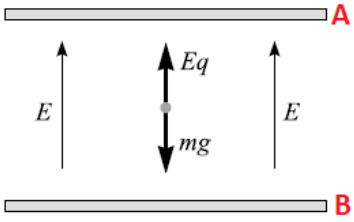
a) -1.5×10^{-18} C b) -3.9×10^{-16} C c) -1.2×10^{-15} C d) -3.9×10^{-15} C

118. A capacitor has a capacitance of 0.093 μF. If the charge on the capacitor is 58 μC, what is the electrical potential difference?

a) 5.4×10^{-12} V b) 1.6×10^{-6} V c) 6.2×10^{2} V d) 5.4×10^{5} V

119. A metal sphere mounted on an insulating rod carries a charge of 6 nC when its potential difference is 200 V higher than its surroundings. What is the capacitance of the capacitor formed by the sphere and its surroundings?

120. **a)** As shown in figure below, a charged particle remains stationary between the two horizontal charged plate. The plate separation is 2 cm and m = 4×10^{-13} kg and |q| = 2.4×10^{-18} C for the particle. Determine the sign of plate A and B?

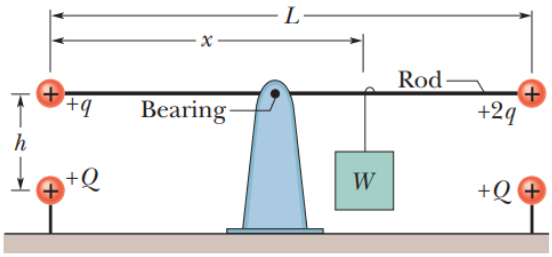


b) Is q positive or negative? Justify your answer.

c) Calculate the electric field between these two plates?

d) Find the potential difference between the plates.

121. Figure below shows a long, nonconducting, massless rod of length L, pivoted at its center and balanced with a block of weight W at a distance x from the left end. At the left and right ends of the rod are attached small conducting spheres with positive charges q and 2q, respectively. A distance h directly beneath each of these spheres is a fixed sphere with positive charge Q.



a. Find the distance x when the rod is horizontal and balanced in terms of L, W, Q and h.

b. What value should h have so that the rod exerts no vertical force on the bearing when the rod is horizontal and balanced?